Atchison

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Capital Market Assumptions

Overview

Capital markets assumptions are the expected returns, standard deviations, and correlation estimates that represent **the long-term risk/return forecasts for various asset classes**. Atchison Consultants (Atchison/We) use these values to assess portfolio risk, assist clients in portfolio construction, construct our own asset allocation models and create Monte Carlo simulation (see Glossary of Terms) inputs for portfolio wealth forecasts.

Atchison's approach to estimating capital markets assumptions and constructing asset allocation models is based on the following general assumptions:

- The global capital markets are largely efficient in the long run, where the efficiency of the markets is measured by the Capital Asset Pricing Model (CAPM), (see Glossary of Terms).
- While the global capital markets are efficient in the long run, there might exist identifiable shorter-term inefficiencies in the capital markets.

Atchison's capital market assumptions construction process is based on using statistical techniques to combine information coming from three sources: theory, researcher views (e.g. forecasts by recognized economic analysts or our own views into future returns of equity and fixed income asset classes), and historical data. The process consists of the steps detailed in the next sections.

In addition, Atchison also provide quarterly tactical asset allocation (TAA) views based on a 12-18 month forecast to capitalise on **shorter term** opportunistic movements in financial markets, based on a regime signal model. Providing flexibility to compliment the more rigid approach that comes with strategic asset allocation.

Our Process

Asset allocation analysis is driven by longer term 10-year market forecasts for each asset class. These forecasts are updated annually. In the event of significant financial economic or market events occurring revisions will be made during the year during the year. Forecasts for asset classes are inputted to our mean-variance optimizer (MVO) (see Glossary of Terms).

The forecast process starts from the analysis of the global economy, followed by the development of the view on the global economy and capital markets. The implications of this view are translated into an outlook for financial and capital markets and each asset class through development of prospective scenarios of a series of forecast returns. These forecasts are for 10 years.

Scenarios reflect the critical variables which will influence the direction and magnitude of movement in investment markets. In approaching our long-term forecasts, we consider the following:

- Potential growth rates (productivity, population growth)
- Growth and inflation assumptions
- Key imbalances in the global economy and in specific markets and regions
- Key transitional themes such as China, climate change, inflation
- Long term interest rate assumptions

Forecasts of returns are prepared using long-term perspectives. Historical data provides evidence of the attributes of each of the asset classes. Return projections are prepared which reflect the broad economic conditions and likely impact on capital and financial market conditions. Focus on extreme conditions such as inflation in goods and services or assets. Fiscal and monetary policies in economies which bias valuations in investment capital markets are key factors.

Building Blocks

We utilise a component part approach to building up our capital market assumptions.

Rather than use historical long-term Earnings Per Share (EPS) growth or a risk-premia approach we can generate more specific projections for EPS growth by breaking down the components of EPS growth such as top-line sales growth, changes to margins and dilution/buybacks.

A summary of key considerations is provided below:

Revenue and Margins
Estimates largely reflect top line sales growth which, in turn, is a function of nominal Growth Domestic Product (GDP) growth in the regions in which the market is exposed.

For Profit margins we assume a partial mean reversion scenario over the longer term. Wages share had been low and declining for number of years, although as inflation rises and political imperatives support an increasing focus on real wages growth. Profit margins are considered relative to long term averages.

Valuation Adjustments
The valuation starting point of an asset has a significant impact on longer term return expectations, particularly when the other drivers of returns are also low. Our approach to valuations is to adopt a range of valuation models and approaches but also to consider the broad investment regime, rather than adopt a purely mean-reversion approach.

Dilution and Buybacks
Our building blocks approach allows for the impact of dilution (new share issuance for existing and new companies) and buybacks on EPS growth. Over the long term several studies have indicated the level of net dilution to be around 1-2% per annum over the long term. In emerging markets, the level of dilution has been higher, reflecting the requirement for growth and expansion.

The level of buybacks has been significant in the US and in some major markets and we project this to remain the case over the medium term, providing an offset to the level of dilution. In emerging markets buybacks are insufficient to come close to offsetting new issuance, creating a drag on EPS growth.

Sovereign Bond Markets
Sovereign bond market long-term prospective returns are largely a function of current yields with the yield on a 10-year bond today providing a relatively good forecast of subsequent 10-year returns. However, we adopt the building blocks approach in projecting bond market returns. We calculate an average yield over the period based and then project views regarding shifts in the yield curve over the forecast period.

A roll return is considered that represents the return from “riding” the yield curve, the steeper the yield curve the higher the roll return. The valuation impact is calculated by the expected changes in the bond yield multiplied by the duration of the bond index.

Credit
We treat credit as a separate asset classes as it offers different risk/return characteristics over the cycle compared with bonds, particularly in more extreme economic scenarios. Returns from credit portfolios are a function of yield, credit loss, roll-down and some valuation adjustments.

Relative spreads are considered across the rating categories. Levels are compared against economic cycle and expected default/recovery rates to consider the level of credit risk compensation offered by markets. Credit spreads mean revert over time as credit conditions change.

Private Assets
Private assets that are infrequently valued, such as property, infrastructure and private equity, have historical records that are particularly stable or smooth in nature when compared to their listed mark-to-market equivalents. Serial correlation analysis

provides an adjustment factor to a smoothed data series. For example, serial correlation assessment of the property asset class typically increases the volatility outcome of a historical series by 3x-5x. This effect is considered when developing the volatility forecasts for private assets. It is noted that the industry standard for historical portfolio analysis is to not adjust the actual volatility of return outcomes of private assets.

Correlations

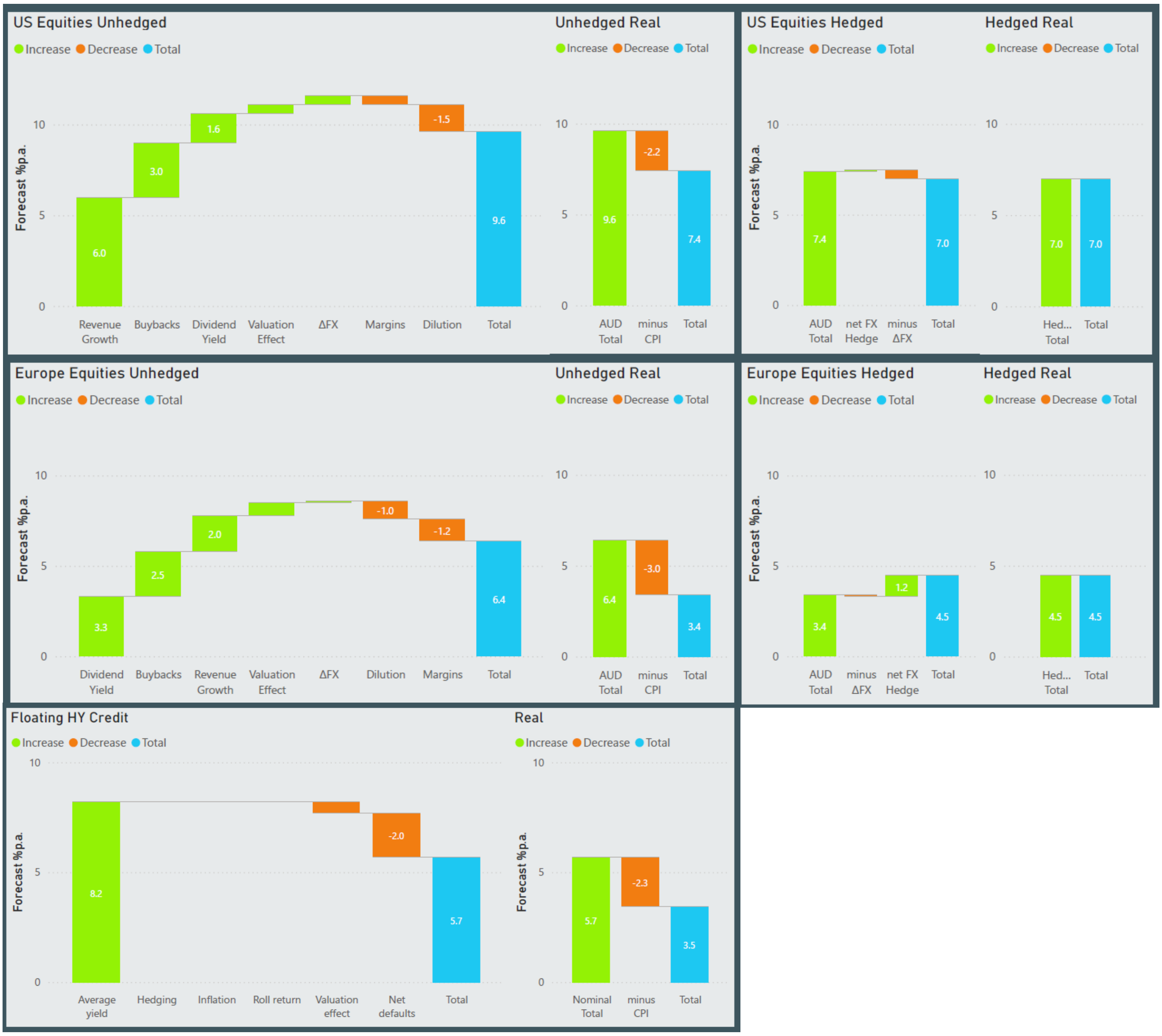
Correlation of returns (see Glossary of Terms) are projected using the preceding five-year historical period, based on the experience of correlations shifting over time and analysis which concludes that five-year historical correlations provide a clear indication of the medium-term future.

Real interest rates are fundamental determinants of fixed interest returns.

Expected volatility or returns is informed by an assessment of historical 15-year volatility of return outcomes. Adjustment is made on a qualitative basis with assessment considering whether a reversion of the level of recent stability or instability is expected.

Worked Examples

Below provides worked examples of our US and European Equity forecasts compared to Atchison's Investment Grade Credit forecasts. Components are selected per asset class to provide relevance of primary drivers of each asset class forecast.



Asset Classes

Below provides a list of the standard asset class forecasts we develop.

All asset class forecasts are available on a nominal or real basis, and global asset forecasts are provided on a hedged or unhedged basis.

US Equities
Europe Equities
Japan Equities
Asia (ex-Japan) Equities
EM Equities
Developed Markets
Global Small Cap
Australian Equities
Australian Mid Cap
Australian Small Cap
A-REITs
G-REITs
Australian Listed Infrastructure
Global Listed Infrastructure
Australian Direct Property
Australian Direct Infrastructure
Commodities

Alternatives - Growth Liquid
Alternatives - Growth Illiquid
Alternatives - Defensive Liquid
Private Equity
Australian Fixed Interest
International Fixed Interest
Floating Government Bonds
Inflation Linked Government Bonds
Emerging Market Debt
Fixed Investment Grade Credit
Floating Investment Grade Credit
Fixed High Yield Credit
Floating High Yield Credit
Private Credit
Hybrids
Term Deposits/ Mortgages
Cash

TAA Signal Model

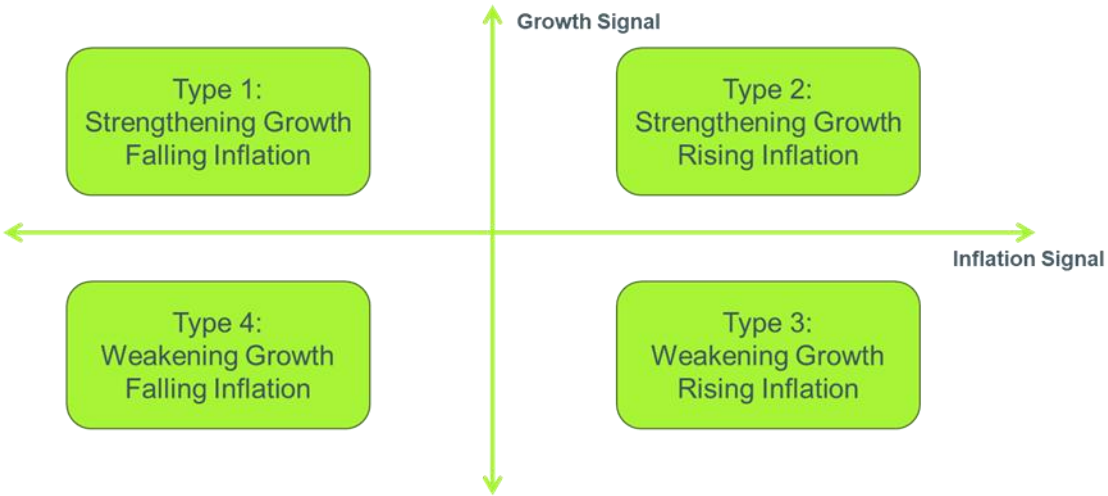
Atchison provide shorter term views on tactical asset allocation (TAA) tilts on a quarterly basis, constructed on a 12-18 month forecast view.

Extremes in markets either of optimism or pessimism is the key focus of analysis. Extremes may persist for extended periods, with qualitative analysis seeking to identify opportunities for relative value or improved risk management for portfolios. A regime signal model provides the core quantitative analysis that underpins our view of changes in trends of economies. Details are provided below:

Overview of Regime Signal Model

Using proprietary research, Atchison has constructed a signal model to identify key current economic conditions on a Country-by-Country basis that then provides predictive power regarding expected forward return profiles.

Atchison's framework creates four possible regimes based on two key metrics: growth and inflation. To minimise the influence of noisy data, these economic conditions must persist for a minimum defined period before they are confirmed as a regime.



The intent is to provide repeatable forward-looking indicators as to the relative attractiveness of major global equity markets – as well as assist in the decision regarding timing of Active management decisions – whether expressed via use of Country or Industry specific ETFs or via selection of Active Managers.

Initial analysis focused on US equities and US economic conditions given the availability of long-term S&P indices and leading indicator datasets back to mid-1950s. Subsequent research has broadening to other countries, styles & industry segments.

To illustrate the relationship between market returns and economic regimes, we overlay the rebased S&P 500 index level onto the economic regimes, with the S&P 500 rebased to January 1958. Aggregation of return profiles by economic regime signal, strongly supports the case that the ability to identify the current trend utilising leading indicators is a major differentiator for investors equity portfolio outcomes.

The return analysis below accounts for lag in underlying collection of datapoints in an attempt to fairly analyse the predictive power of the basket of underlying indicators used.

Table 1: S&P 500 Returns by Signal Regime – 1958 to 2025

67 Years (1958 to 2025)	# Mths	% Mths	Av p.a. Return
Type 1: Strengthening Growth & Falling Inflation	252	32%	19.5%
Type 2: Strengthening Growth & Rising Inflation	156	20%	17.5%
Type 3: Weakening Growth & Falling Inflation	178	22%	7.5%
Type 4: Weakening Growth & Rising Inflation	209	26%	-1.8%
Total	**795**	**100%**	**10.5%**

Historically, the S&P 500 exhibited the highest absolute returns and risk-adjusted returns in the Rising Growth and Falling Inflation regime, followed by the Rising Growth and Rising Inflation regime. Conversely, the S&P 500 experienced significantly lower—or even negative—returns in the Slowing Growth and Falling Inflation and the Slowing Growth and Rising Inflation regimes.

Example of Regime Analysis Applied on Factor Strategies

The following analysis builds on the above general market regime analysis presented above, by analysing whether the use of factor/style-based investment strategies can consistently add value or not above general market returns – and importantly under what economic regimes they are more likely to add or detract value for investors wanting to utilise style heavy active management strategies.

Table 2: Regime Analysis - Style Factor

Outcome	Rising Growth & Falling Inflation	Rising Growth & Falling Inflation	Slowing Growth & Falling Inflation	Slowing Growth & Falling Inflation	All Periods
Best Return	Growth	Momentum	Quality	Low Vol	Quality
	Value	Growth	Low Vol	Yield	Momentum
	Size	**S&P 500**	Momentum	Quality	Growth
	Momentum	Quality	**S&P 500**	Value	Size
	S&P 500	Size	Growth	Size	Value
	Quality	Value	Size	**S&P 500**	**S&P 500**
	Yield	Yield	Yield	Momentum	Low Vol
Worse Return	Low Vol	Low Vol	Value	Growth	Yield

Building further, the following analysis assesses the impact on risk-adjusted returns for investors utilising factor/style-based investment strategies, taking into consideration whether excess or reduced volatility changes the equation for each style and regime.

Table 3: Ranking Factor/Style Strategy Risk-Adjusted Performance – 20 Years

Outcome	Rising Growth & Falling Inflation	Rising Growth & Falling Inflation	Slowing Growth & Falling Inflation	Slowing Growth & Falling Inflation	All Periods
Best Risk Adjusted Return	Quality	Momentum	Low Vol	Low Vol	Quality
	Momentum	**S&P 500**	Quality	Yield	Low Vol
	S&P 500	Quality	Momentum	Quality	Momentum
	Growth	Growth	**S&P 500**	Value	**S&P 500**
	Size	Size	Growth	Size	Size
	Value	Value	Size	**S&P 500**	Yield

Worse Risk	Low Vol	Yield	Yield	Growth	Growth
Adjusted Return	Yield	Low Vol	Value	Momentum	Value

Similar analysis has been prepared on across matrices of Countries and Industry Sectors.

Glossary of Terms

Capital Asset Pricing Model (CAPM). Used to determine a theoretically appropriate required rate of return of an asset, if that asset is to be added to an already well-diversified portfolio, given that asset's non-diversifiable risk. The model takes into account the asset's sensitivity to non-diversifiable risk (also known as systematic risk or market risk), often represented by the quantity beta (ß) in the financial industry, as well as the expected return of the market and the expected return of a theoretical risk-free asset. The CAP was independently authored by Jack Treynor (1961,1962), William Sharpe (1964), John Lintner (1965), and Jan Mossin (1966).

Correlation, A statistical measure of how two securities move in relation to each other. Perfect positive correlation (a coefficient of +1) implies that as one security moves, either up or down, the other security will always move in the same direction. Perfect negative correlation (a coefficient of -1) means that if one security moves up or down, the negatively correlated security will always move in the opposite direction. If two securities are uncorrelated, the movement in one security does not imply a linear movement up or down in the other security.

Mean-variance optimization. A method to select portfolio weights (allocation to assets) aiming to maximise returns for a given level of risk or minimise risk for a given level of return.

Monte Carlo simulation. Is a way to model the probability of different outcomes in a process that cannot easily be predicted due to intervention of random variables. The technique is used to understand the impact of risk and uncertainty.

Standard deviation. A statistical measure of dispersion of the observed return, which depicts how widely a stock, or portfolio's returns varied over a certain period-of-time. When a stock or portfolio has a high standard deviation, the predicted range of performance is wide, implying greater volatility.

2025 Capital Markets Assumptions

AssetClassName	2025		2024		As of 31 December 2024	
	Forecast 10yr Return	Forecast 10yr Volatility	Forecast 10yr Return	Forecast 10yr Volatility	Historical 10yr Return	Historical 10yr Volatility
Australian Equities	7.45%	14.00%	8.75%	14.50%	8.51%	13.78%
Australian Equities - Small Cap	7.60%	15.00%	10.00%	16.50%	8.87%	15.76%
International Equities - Unhedged	8.89%	13.08%	9.12%	13.08%	13.16%	11.25%
International Equities - Hedged	8.87%	14.08%	9.10%	14.08%	10.26%	14.14%
International Equities - US	9.60%	16.00%	9.50%	19.80%	15.67%	12.02%
International Equities - Emerging	6.70%	19.00%	6.30%	20.00%	6.57%	11.13%
International Equities - Small Cap	10.10%	18.00%	6.00%	19.00%	10.54%	12.86%
Private Equity	13.00%	21.00%	13.00%	21.50%	3.52%	10.66%
AREITs	6.50%	19.00%	7.00%	20.00%	8.40%	20.53%
GREITs	6.45%	17.00%	6.45%	18.00%	5.62%	11.98%
Australian Direct Property	4.50%	7.00%	4.75%	7.00%	5.38%	3.95%
Australian Direct Infrastructure	5.25%	9.00%	5.50%	8.50%	5.38%	6.82%
Australian Listed Infrastructure	5.75%	15.00%	6.00%	15.70%	9.72%	15.21%
Commodity	5.00%	17.00%	5.00%	16.50%	4.14%	12.23%
Alternatives - Growth Liquid	5.50%	8.00%	6.00%	7.50%	5.23%	8.64%
Alternatives - Growth Illiquid	6.00%	6.00%	5.50%	5.30%	3.57%	10.67%
Australian Fixed Interest	4.70%	3.25%	4.85%	3.50%	1.97%	4.47%
International Fixed Interest	4.45%	3.00%	4.50%	3.20%	1.85%	4.04%
Floating Government Bonds	4.35%	3.50%	4.20%	3.50%	1.72%	5.04%
Inflation Linked Government Bonds	4.30%	3.25%	3.25%	3.75%	2.47%	6.34%
Fixed Investment Grade Credit	5.00%	4.00%	5.55%	3.25%	2.16%	5.56%
Floating Investment Grade Credit	5.15%	3.00%	5.15%	4.00%	2.85%	0.70%
Fixed High Yield Credit	6.10%	9.00%	6.65%	7.50%	4.68%	8.03%
Floating High Yield Credit	5.70%	6.00%	5.90%	6.00%	4.46%	4.69%
Private Credit	7.50%	9.20%	9.90%	8.00%	9.01%	2.86%
Hybrids	5.75%	7.00%	5.80%	5.00%	5.16%	3.19%
Term Deposits/Mortgages	4.35%	1.50%	3.75%	1.50%	3.03%	2.94%
Cash	3.00%	0.50%	3.00%	0.50%	1.95%	0.39%
Alternatives - Defensive	5.00%	5.00%	6.50%	5.00%	9.01%	2.86%
Gold	5.00%	16.50%	5.00%	16.50%	11.05%	14.39%
CPI	2.75%	0.50%	2.75%	0.50%	2.72%	1.35%

Disclosure

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